GOOGLE PLAY STORE APP ANALYTICS

Tableau Project Report

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Data Preparation • Exploratory Analysis • Interactive Dashboards • Business Recommendations

1. Executive Summary

This project is designed to analyze Google Play Store application data in Tableau to understand app popularity, user ratings, review engagement, pricing, app size, and user sentiment. The planned output is an interactive dashboard that helps app developers, product managers, marketing teams, and senior management explore patterns in app performance and identify areas for further investigation.

The report documents the proposed workflow, including data connection, data quality checks, calculated fields, visualization development, dashboard interactions, and publishing. It is a project plan and implementation guide; it does not claim measured findings because no validated Tableau outputs or computed results were provided.

2. Project Background

The project uses Google Play Store data containing application attributes such as category, rating, number of reviews, size, installs, type, price, content rating, genre, and last-updated date. A separate user-review dataset can support sentiment analysis where review text and sentiment fields are available.

3. Problem Statement

The objective is to investigate patterns that may be associated with app success, including the relationship between ratings and installs, differences across categories, the free-versus-paid app split, app size, pricing, and user feedback. The analysis should support evidence-based product and marketing decisions without assuming that observed relationships prove causation.

4. Project Objectives

- Measure app counts, average ratings, review counts, and reported install figures.
- Compare application performance across categories, genres, content ratings, and pricing types.
- Explore relationships between app size, installs, reviews, ratings, and price.
- Analyze available user-review sentiment to identify possible feedback themes.
- Build interactive Tableau dashboards with filters and drill-down actions.
- Translate validated findings into practical recommendations for stakeholders.

5. Stakeholders

Stakeholder	Potential use of the analysis
App developers	Identify areas for app quality, usability, and performance improvement.
Product managers	Compare category performance and prioritize product opportunities.
Marketing team	Understand reported popularity and audience patterns.
Senior management	Review portfolio-level performance indicators and strategic opportunities.
App users and partners	Indirectly benefit from better-informed product decisions and user experience improvements.

6. Dataset Description

Dataset	Key fields / purpose
googleplaystore.csv	App, Category, Rating, Reviews, Size, Installs, Type, Price, Content Rating, Genres, Last Updated, Current Ver, Android Ver.
googleplaystore_user_reviews.csv	Typically includes App, Translated_Review, Sentiment, Sentiment_Polarity, and Sentiment_Subjectivity; verify exact fields in the supplied file.

The app dataset is primarily app-level, while the reviews dataset may contain multiple rows per app. This difference in granularity must be considered before combining the sources; otherwise, app-level counts or measures can be duplicated.

7. Tools and Technologies

- Tableau Desktop or Tableau Public for data visualization and dashboard development.
- CSV files as source data.
- Tableau calculated fields for data type conversion and derived metrics.
- Optional text preprocessing for deeper review-theme analysis.

8. Data Preparation and Quality Checks

1. **Connect data:** Import the main CSV through the Text File connector. Add the review CSV as a second data source or related table when required.
2. **Inspect missing values:** Review nulls in Rating, Size, Price, Installs, and review text. Keep missing ratings as null rather than treating them as zero.
3. **Convert installs:** Remove commas and plus signs from reported install strings, then convert to a numeric field. Treat the result as reported install thresholds, not exact download counts.
4. **Convert reviews:** Convert the Reviews field to a numeric type after checking for invalid or missing values.
5. **Convert price:** Remove currency symbols only after verifying the source format and maintaining a consistent currency.
6. **Normalize app size:** Convert values in megabytes to numeric values. Handle kilobytes, missing values, and 'Varies with device' separately.
7. **Parse dates:** Confirm that Last Updated is a valid date and parse it using the source date format.
8. **Check duplicates:** Inspect repeated app names and records before removing anything. Repeated names are not automatically duplicate records.
9. **Validate relationship:** When relating reviews to apps, use App as a candidate key and verify that measures are not inflated by the different levels of detail.

9. Calculated Fields

The following examples can be created from the Data pane using Create Calculated Field. Confirm field names and data types in the actual workbook before using them.

9.1 Installs Numeric

Purpose: convert values such as 1,000+ into a numeric reported-install value.

```
INT(REPLACE(REPLACE([Installs], "+", ""), ",", ""))
```

9.2 Reviews Numeric

```
INT([Reviews])
```

9.3 Price Numeric

```
FLOAT(REPLACE([Price], "$", ""))
```

Use only if the source price values contain a dollar sign. Review invalid values before applying the calculation.

9.4 App Size MB

```
IF CONTAINS([Size], "M") THEN FLOAT(REPLACE([Size], "M", "")) ELSE NULL END
```

This basic calculation handles MB values only. Values in KB and non-numeric size descriptions need additional handling.

9.5 Rating Group

```
IF ISNULL([Rating]) THEN "Not Rated"
ELSEIF [Rating] >= 4.5 THEN "Excellent"
ELSEIF [Rating] >= 4.0 THEN "Good"
ELSEIF [Rating] >= 3.0 THEN "Average"
ELSE "Below Average" END
```

9.6 App Type

```
IF [Type] = "Free" THEN "Free"
ELSEIF [Type] = "Paid" THEN "Paid"
ELSE "Unknown" END
```

9.7 Install Range

```
IF ISNULL([Installs Numeric]) THEN "Unknown"
ELSEIF [Installs Numeric] < 1000 THEN "Below 1K"
ELSEIF [Installs Numeric] < 100000 THEN "1K–99K"
ELSEIF [Installs Numeric] < 1000000 THEN "100K–999K"
ELSEIF [Installs Numeric] < 10000000 THEN "1M–9.9M"
ELSE "10M+" END
```

10. KPI Definitions

KPI	Suggested definition	Interpretation / caution
Total Apps	COUNTD([App])	Counts distinct app names, not necessarily unique source records.
Average Rating	AVG([Rating])	Null ratings are excluded by the average.
Reported Installs	SUM([Installs Numeric])	Install values may be thresholds, so label as reported installs.
Total Reviews	SUM([Reviews Numeric])	Review count is engagement volume, not a satisfaction score.
Apps Rated 4.0+	COUNTD(IF [Rating] >= 4 THEN [App] END)	Validate app uniqueness and rating completeness.
Free/Paid App Count	COUNTD([App]) split by [App Type]	Ensure unknown Type values are handled consistently.

11. Worksheet Development Plan

Worksheet	Chart type	Purpose
Total Apps	KPI card	Show distinct app count.
Average Rating	KPI card	Show mean rating on the 0–5 scale.
Sum of Reported Installs	KPI card	Show summed reported install figures with a clear caveat.
Apps by Category	Horizontal bar chart	Compare the number of apps across categories.
Top 10 Categories by Installs	Ranked bar chart	Compare categories by sum of reported installs.
Average Rating by Category	Ranked bar chart	Compare category-level average ratings.
Free vs Paid Apps	Bar chart	Compare counts by app type.
App Size vs Rating	Scatter plot	Explore the association between app size and rating.
Reviews vs Installs	Scatter plot	Explore the association between review volume and reported installs.
Ratings by Content Rating	Bar chart / box plot	Compare ratings across content-rating groups.
Review Sentiment	Bar chart	Count records by available sentiment category.
Negative Reviews by App	Ranked bar chart	Find apps with the largest number of negative review records.

12. Dashboard Design

12.1 Dashboard 1: Google Play Store Overview

- Top row: Total Apps, Average Rating, and Sum of Reported Installs KPI cards.
- Middle row: Apps by Category and Top 10 Categories by Installs.
- Bottom row: Average Rating by Category and Free vs Paid Apps.
- Filters: Category, App Type, Content Rating, Rating Group, and Install Range.

12.2 Dashboard 2: App Performance

- Average rating by category and reported installs by category.
- Reviews versus reported installs and app size versus rating scatter plots.
- Rating distribution by content rating and free-versus-paid comparisons.
- Use tooltips and filters to help users explore subsets of apps.

12.3 Dashboard 3: Review Sentiment

- Counts of positive, neutral, and negative review records, where available.
- Negative review counts by app and sentiment polarity distribution.
- Sentiment by category and frequent negative-review themes if text processing is completed.
- Clearly state that review records may not represent all users or all reviews.

13. Interactivity and User Experience

- Add relevant filters and apply them to appropriate worksheets.
- Use a dashboard filter action so selecting a category filters related charts.
- Create informative tooltips showing app name, category, rating, review count, and reported installs.
- Use consistent formatting, readable labels, and a limited color palette.
- Test filter behavior and totals before publishing.

14. Interpretation and Limitations

- The report describes an analysis plan; no actual results are stated because calculated Tableau outputs were not supplied.
- Reported install figures are threshold-like values and should not be described as exact download totals.
- Average ratings can be misleading when groups have very different numbers of rated apps.
- Correlation between installs, reviews, app size, price, and rating does not prove causation.
- App-level data and review-level data have different granularities. Combining them without validation can duplicate app-level metrics.
- Missing values and inconsistent size formats can affect results; document the chosen cleaning rules.
- Sentiment labels and polarity scores depend on the method used to generate them and may not fully capture context or sarcasm.

15. Recommendations Framework

After building the dashboards, use the validated results to develop recommendations. Examples of decision rules include:

- Prioritize further review of categories with strong average ratings and meaningful reported install volume.
- Investigate categories with high reported installs but lower average ratings for possible usability or quality issues.
- Compare free and paid apps using both rating and review engagement rather than relying on one metric.
- Use negative review themes to identify recurring product issues, then validate those themes with product teams.
- Treat app size as a factor to investigate, not as a direct cause of app performance.

16. Conclusion

This project provides a structured Tableau workflow for analyzing Google Play Store app performance and user feedback. It covers data preparation, KPI creation, category comparisons, pricing analysis, scatter plots, sentiment analysis, and interactive dashboard design. Once the workbook is built and the results are validated, the dashboards can help stakeholders explore app performance patterns and formulate evidence-based product and marketing recommendations.